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TITLE:

SUPERMARKET SALES

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CHAPTER 1: INTRODUCTION

1.1 Background of Study

Supermarket sales have become an increasingly significant area of analysis in recent years. As essential businesses, supermarkets fulfil a wide range of consumer needs, offering products and services that are central to daily life. Their sales performance serves as a barometer for understanding consumer behaviour, economic conditions, and the operational efficiency of the retail sector. By providing access to affordable and diverse goods, supermarkets play a pivotal role in supporting the socio-economic well-being of communities. They also serve as indicators of regional economic development, employment creation, and business innovation (Chen et al., 2019). Analysing sales trends is crucial for managers, investors and policymakers as these insights inform critical decisions regarding pricing strategies, inventory control, and marketing efforts.

A variety of factors impact supermarket sales, including demographic characteristics, purchasing habits, payment preferences, product pricing, and seasonal trends (Kotler & Keller, 2016). For instance, shopping behaviours often differ between urban and suburban customers, and sales promotions or discounts can strongly influence consumer spending patterns. Customer satisfaction is another vital aspect for supermarkets striving to retain loyal shoppers and attract new ones. Elements such as product availability, competitive pricing, and ease of payment significantly shape the overall customer experience (Zeithaml et al., 2020). This study seeks to examine supermarket sales data to identify purchasing trends, assess branch performance, and analyse customer satisfaction. The analysis will leverage a publicly available dataset containing variables such as branch location, customer type, product categories, payment methods, and total sales, offering a robust foundation for evaluating supermarket operations.

1.2 Research Objectives

i. To determine if there is a significance relationship between gross income and factors affecting it such as the number of items sold, price per unit sold and 5% or tax.
ii. To determine if there is a significant difference in the median of gross income between gender.
iii. To determine if there is a significant difference in the average of gross income across different product lines.
iv. To determine whether the median rating of products in the dataset is significantly different from a hypothesized value.
v. To determine if there is a significant relationship between total sales and cost of goods sold (COGS).

1.3 Data Description

Variable	Variable Names	Description	Type of Variable
Y	TOTAL	Total Sales	Numeric
X	BRANCH	Branch of supercentre	Categorical
	CITY	Location of supercentre	Categorical
	GENDER	Gender type of customer	Categorical
	PRODUCT_LINE	General item categorization groups	Categorical
	UNIT_PRICE	Unit price for each product	Numeric
	QUANTITY	Number of products purchased by customer	Numeric
	TAX	5% tax fee	Numeric
	PAYMENT	Payment method	Categorical
	COGS	Cost of goods sold	Numeric
	GROSS_INCOME	Gross income	Numeric
	RATING	Customer stratification rating	Numeric

Table 1: List of variables

1.4 Summary of Analysis

No.	Objective	Variables	Method
1.	To predict the value of total sales based on the value unit price and branch.	• Total sales • Branch • Gender • Product Line	Multiple Linear Regression
2.	To determine if there is a significant difference in the median of gross income between gender.	• Gross income • Gender	Wilcoxon Rank Sum Test (Mann-Whitney U Test)
3.	To determine if there is a significant difference in the average gross income across different product lines.	• Gross income • Product line	Kruskal Wallis Test
4.	To determine whether the median rating of products in the dataset is significantly different from a hypothesized value.	• Rating	One Sample Median Test
5.	To determine if there is a significant relationship between total sales and cost of goods sold (COGS).	• Total Sales • Cost of Goods Sold (COGS)	Spearman's Rank Correlation Coefficient

Table 2: Summary of Analysis

CHAPTER 2: METHODOLOGY

2.1 Introduction

This section explains the strategies used to achieve our objective. The original data, which was in csv format in Excel, was transformed into an SAS dataset. As a result, the SAS procedure and data step statement may be used with the data from Kaggle website.

2.2 Data Cleaning

In order to arrange the data, the raw data must be cleaned, either using SAS or manually through excel. It will simplify the data. The data cleansing in this study was done by utilizing label statements to change the names of the variables.

2.3 Data Management

2.3.1 Assigning a libref

A libref was assigned to an SAS data library using the LIBNAME command, and the libref for this study is 'orion'.

LIBNAME declaration in its most general form:

```
LIBNAME libref 'SAS-library';
```

2.3.2 Reading Excel spreadsheets

To build an SAS dataset from the spreadsheets, the IMPORT method was used. The IMPORT statement in basic version is:

```
LIBNAME libref 'SAS-library';
PROC IMPORT DATAFILE = 'external-file-name';
OUT = <libref> 'SAS-data-set'
DBMS = 'identifier';
RUN;
```

2.3.3 Browsing the descriptor portion

The CONTENTS method displays the SAS dataset's description component. The following is found in the description section:

a. General information about the SAS dataset such as dataset name (orion.smsales) and the number of observations (1000 observations).
b. Variable properties (variable name, type, length, format, informat and label).

General form of the CONTENTS procedure:

```
PROC CONTENTS data = 'SAS-data-set';
RUN;
```

2.3.4 Printing selected variables

The VAR statement can be used to specify the order in which variables should be included in the report.

VAR statement in basic form:

```
PROC PRINT data= 'SAS-data-set';
VAR variable(s);
RUN;
```

2.3.5 Defining titles

Some of the SAS outputs were given proper titles, and the title statement was used. The page's titles will show at the top.

The global TITLE statement specifies title lines for SAS output:

```
TITLEn 'text';
```

2.4 Descriptive Statistics

2.4.1 Pie Chart

The PROC GCHART statement was used to generate a chart in SAS programming. The distribution value of a variable, the average value of a variable for different categories, and the total value of a variable for various values may all be displayed using a bar or pie chart. The PIE3D is utilised in this study to illustrate the proportion for the categorical variables of branch, city, gender, product line and payment methods. To illustrate the total sales across different branch and product line, the chart is displayed as graphical output rather than HTML output.

General form of the PROC GCHART statement:

```
PROC GCHART DATA='SAS-data-set';
```

Statement to specify the desired type of chart:

```
PROC GCHART DATA='SAS-data-set';
HBAR3D chart-variable...</options>;
VBAR chart-variable...</options>;
PIE3D chart-variable...</options>;
```

2.4.2 Frequency Report

The SAS data set's category and numerical variables are detailed in the frequency report because it will process every unique value for the specified variables, it is the most appropriate approach to show on categorical variables. By default, the PROC FREQ displays the variables' frequency, percentage, row percentage, and column percentage. The PROC FREQ statement is used to generate a cross tabular frequency report between furnishing status and home price in this study. The research examines all conceivable combinations of the various variables associated with payment methods and total sales.

General form of the PROC FREQ statement:

```
PROC FREQ DATA='SAS-data-set';
TABLES variable1*variable2;
RUN;
```

2.4.3 Summary Statistics

The summary statistics in the SAS data set give information on the numerical variables. Simple descriptive statistics are displayed with the PROC MEANS statement. By default, the PROC MEANS displays the number of rows while ignoring missing data, mean, standard deviation, and variable minimum and maximum values. The PROC MEANS statement is used to create a summary statistic for all sales in this research. The data for the branch were limited to the number of observations, mean, median, and mode of sales among various sales.

General form of the PROC MEANS statement:

```
PROC MEANS DATA='SAS-data-set';
RUN;
```

2.5 Statistical Analysis

2.5.1 Test for Normality

The test for normality is to test whether the data comes from a normal distribution.

General form of the test for normality:

```
PROC UNIVARIATE DATA=SAS-data-set;
        VAR variable(s);
        PROBPLOT/NORMAL (mu=est sigma=est);
RUN;
```

2.5.2 Multiple Linear Regression

The purpose of Multiple Linear Regression is to identify whether several independent variables (IV) are significant predictors of the dependent variable (DV). Besides that, it can be used to describe the relationship between the dependent variable and more than one independent variable(s). MLR is also used to predict the value of a dependent variable based on the value of one or more independent variable.

```
PROC REG DATA=SAS-data-set;
        MODEL variable=variable(s);
       OUTPUT OUT=resids p=yhat r=resid;
RUN;
QUIT;
```

2.5.3 Wilcoxon Rank Sum Test (Mann-Whitney U Test)

The Wilcoxon rank-sum test is a non-parametric alternative to the independent t-test in cases where the data for the variable of interest is not normally distributed. This test is used to determine if there is a statistically significant difference between the medians of two independent groups. Independent groups, also referred to as unpaired or unrelated groups, consist of distinct and separate populations.

```
PROC NPAR1WAY DATA= SAS-data-set;
        CLASS variable(s);
        VAR variable(s);
```

```
RUN;
```

2.5.4 Kruskal Wallis Test

The Kruskal-Wallis Test, an alternate version of ANOVA is a rank-based nonparametric test that can be used to determine if there are statistically significant differences between two or more groups of an independent variable on a continuous or ordinal dependent variable. It is a generalized form of the Mann-Whitney test method since it allows comparison of two or more groups.

```
PROC NPAR1WAY DATA=SAS-data-set WILCOXON;
        CLASS variable(s);
        VAR variable(s);
RUN;
```

2.5.5 One Sample Median Test

The one-sample median test is a non-parametric statistical test used to determine whether the population median significantly differs from a hypothesized value. It is commonly employed when the variable being tested is not normally distributed, serving as a substitute for the one-sample t-test, which assumes normality.

SAS Code:

```
PROC UNIVARIATE DATA=your_data mu0=your_hypothesized_value;
        VAR your_test_variable;
RUN;
```

2.5.6 Spearman's Rank Correlation Coefficient

The Spearman's Rank Correlation Coefficient is the nonparametric version of the Pearson product-moment correlation. Spearman Correlation Coefficient is a descriptive measure used to describe the strength of relationships between two quantitative variables when the variables are not normally distributed. In order to find the plot of correlation matrix, plot option should be used.

```
PROC CORR DATA= SAS-data-set spearman rank;
        VAR variables(s);
        With variable(s);
RUN;

PROC CORR DATA= SAS-data-set spearman
plots=matrix(histogram);
        VAR variable(s);
        WITH variable(s)
RUN;
```

CHAPTER 3: RESULT AND ANALYSIS

3.1 Introduction

This section discusses the results and analysis of this study. The data was analysed by using the SAS program, which is SAS Enterprise Guide. The SAS code and some of the outputs were attached in the appendices. The objective is to analyse the descriptive statistics and summarize key features to gain insights into the sales patterns.

3.2 Descriptive Statistics

3.2.1 Summary statistics

In summary statistics, the MEAN procedure (refer **Appendix 5**) was used to display the population (N), mean, standard deviation, minimum and maximum value.

The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
Unit_price	Unit_price	1000	55.6721	26.4946	10.0800	99.9600
Quantity	Quantity	1000	5.5100	2.9234	1.0000	10.0000
Total	Total	1000	322.9667	245.8853	10.6785	1042.6500
gross_income	gross_income	1000	15.3794	11.7088	0.5085	49.6500

Table 3: The Summary Statistics Table for Unit Price, Quantity, Total and Gross Income

Table 3 shows the mean value for Total has the highest mean value of 322.9667, with a minimum of 10.6785 and a maximum of 1042.6500. Quantity has the lowest mean value of 5.5100, ranging from a minimum of 1.0000 to a maximum of 10.0000. Unit price has a mean of 55.6721, with values ranging from 10.0800 to 99.9600. Gross income has a mean of 15.3794, with a minimum 0.5085 and a maximum of 49.6500.

3.2.2 One-Way Frequency Report

In a one-way frequency report, the FREQ procedure (refer **Appendix 6**) was used to display the frequency counts for the product line categories available in the dataset. The TABLE statement was used to limit the variables included in the frequency counts.

The FREQ Procedure

Product_line				
Product_line	**Frequency**	**Percent**	**Cumulative Frequency**	**Cumulative Percent**
Electronic accessories	170	17.00	170	17.00
Fashion accessories	178	17.80	348	34.80

Food and beverages	174	17.40	522	52.20
Health and beauty	152	15.20	674	67.40
Home and lifestyle	160	16.00	834	83.40
Sports and travel	166	16.60	1000	100.00

Table 4: The One-Way Frequency Table for Product Line

Based on Table 4, the Product Line “Fashion accessories” has the highest frequency with 178, accounting for 17.8% of total sales. Meanwhile, the Product Line "Health and beauty” has the lowest frequency, with 152, contributing only 15.2% of total sales.

3.2.3 Pie Chart

Pie chart is being used to visually represent the percentage distribution of supermarket sales based on four key factors Branch, City, Gender, and Payment Method. These charts provide an overview of how sales are distributed across different branches and cities, the proportion of sales by customer gender, and the breakdown of preferred payment methods (refer **Appendix 7**) for the SAS code.

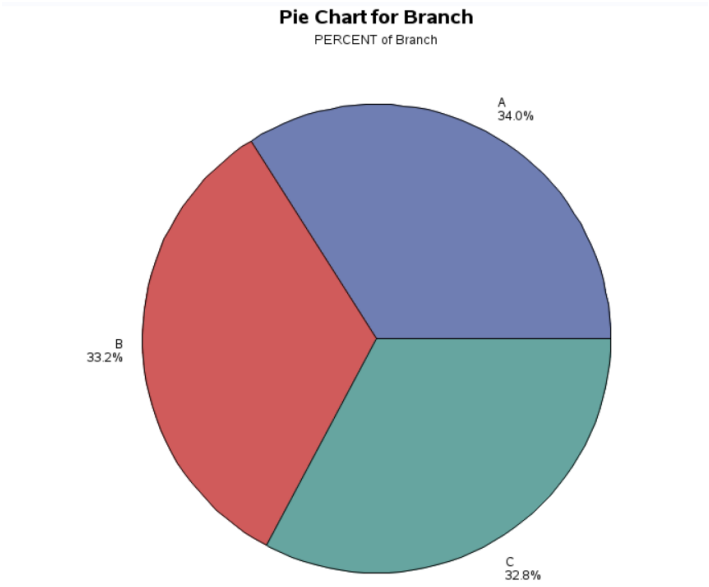


Figure 1: Pie Chart for Branch

Based on Figure 1, pie chart shows how sales are divided among the three supermarket branches A, B, and C. Branch A has the largest share at 34.0%, while Branch B comes next with 33.2%, and Branch C follows closely at 32.8%. The percentages are quite similar, with less than a 2% gap between the highest and lowest values, suggesting that sales are fairly evenly split across the branches.

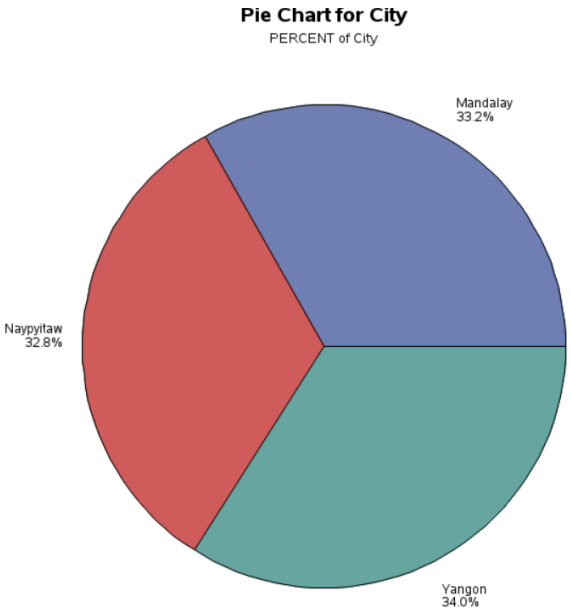


Figure 2: Pie Chart for City

Based on Figure 2, pie chart represents the percentage distribution of three cities, Yangon, Mandalay, and Naypyitaw. Yangon has the highest percentage at 34.0%, followed by Mandalay at 33.2%. Naypyitaw accounts for 32.8%, making it the lowest. The distribution is relatively even, with all three cities having nearly equal representation.

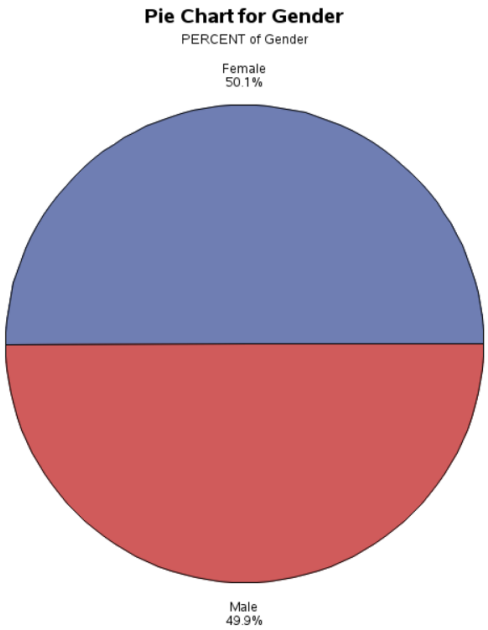


Figure 3: Pie Chart for Gender

Based on Figure 3, pie chart shows the gender distribution in the dataset. Females make up 50.1% of the total, while males represent 49.9%. This indicates that the dataset has an almost equal number of males and females, with females being slightly more.

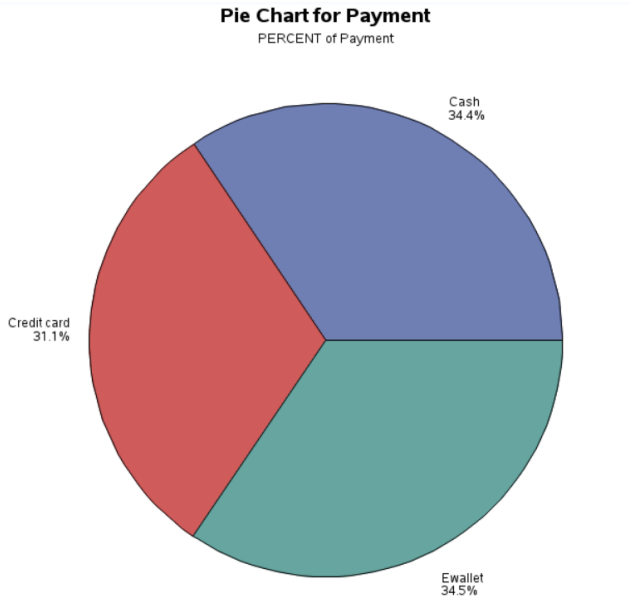


Figure 4: Pie Chart for Payment

Based on Figure 4, pie chart represents the percentage of payment methods used in the dataset. Ewallet is the most preferred method, accounting for 34.5%. Cash follows closely at 34.4%, making it the second most used method. Credit card payments make up 31.1%, which is the least used method. The data shows that ewallet and cash are almost equally popular, with credit card usage slightly lower.

3.3 Inferential Statistics

3.3.1 Normality Test

Refer **Appendix 8** for the SAS code.

a. Quantity

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.932585	$Pr < W$	<0.0001

Table 5: Normality Test for Quantity

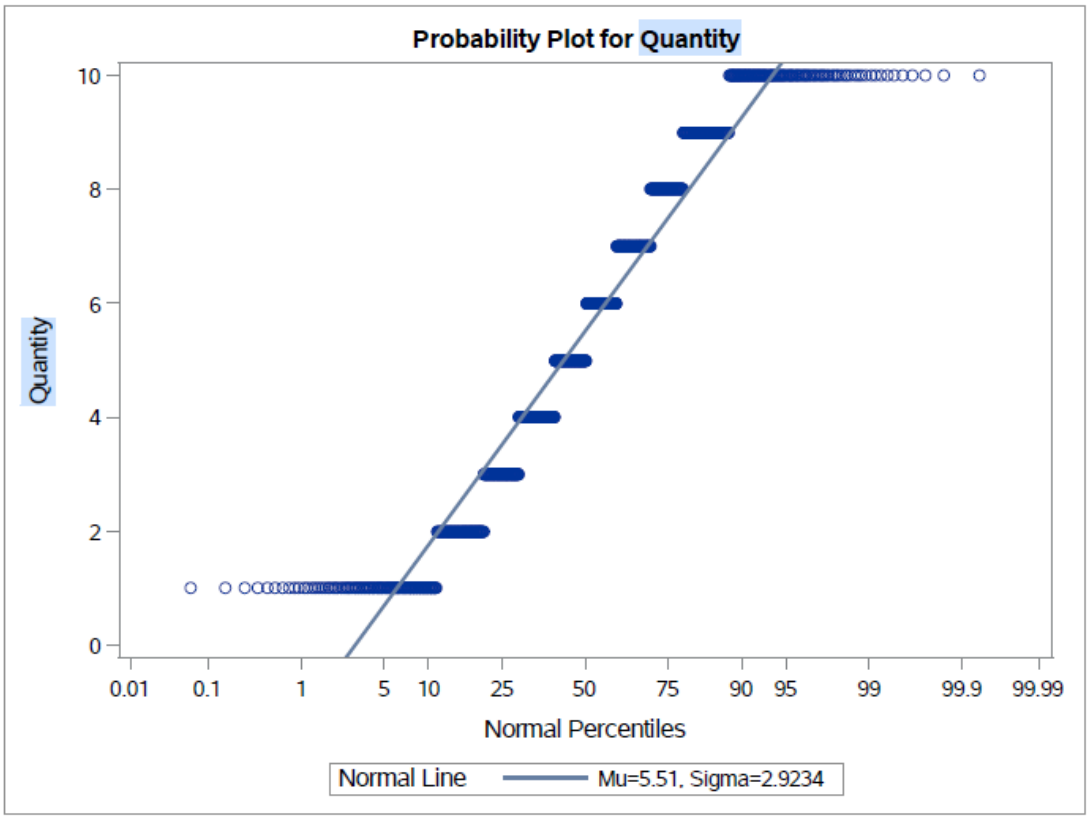


Figure 5: Normal P-P Plot of Quantity

Hypothesis:

H_0 = The quantity is normally distributed.

H_1 = The quantity is not normally distributed.

$\alpha = 0.05$

$p - value = 0.0001$

Decision: Reject H_0 since the $p - value = 0.0001 < \alpha = 0.05$

Conclusion: The quantity is not normally distributed.

Interpretation:

Table 5 shows that there is enough evidence to conclude that quantity of item sold is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 5 which shows that the distribution is not normal in the probability plot of quantity.

b. Rating

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.958201	$Pr < W$	<0.0001

Table 6: Normality Test for Rating

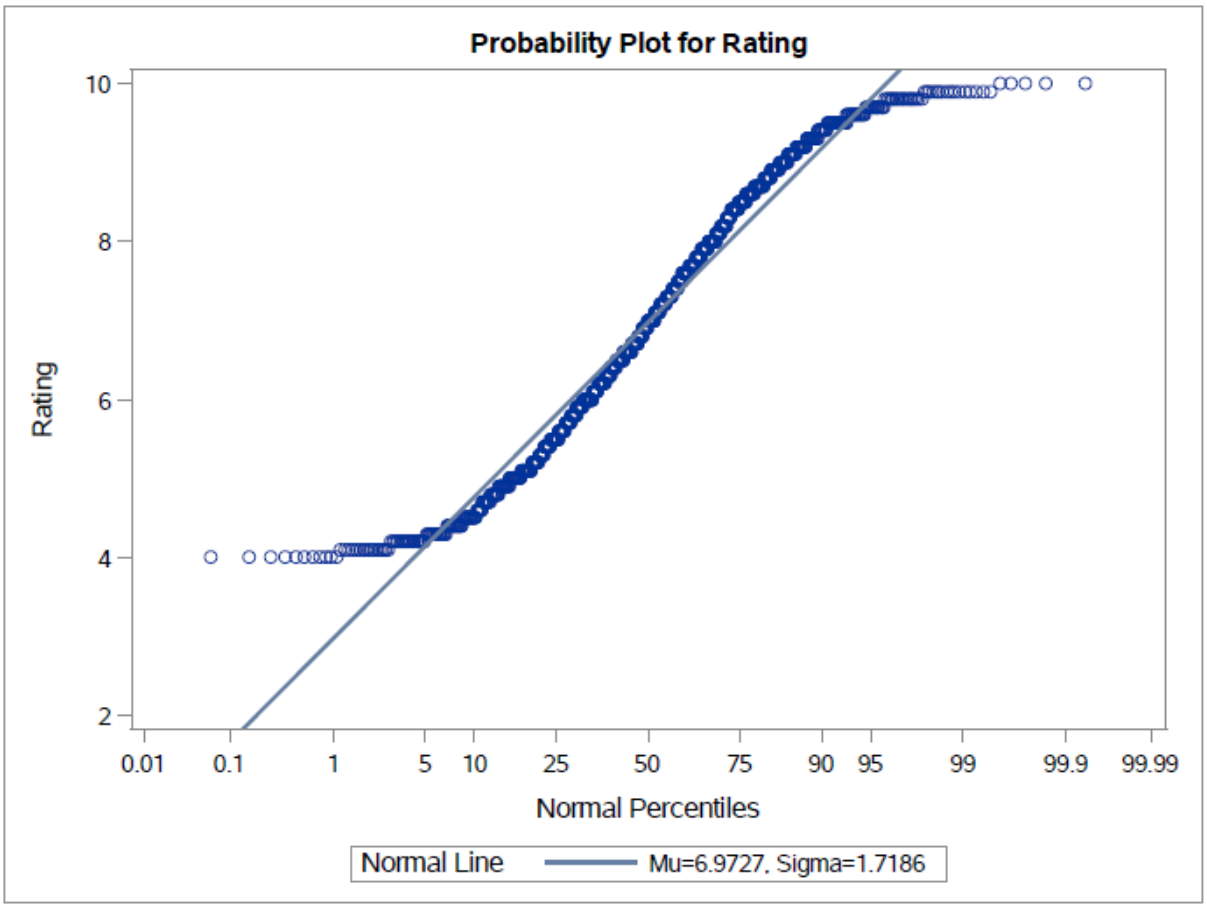


Figure 6: Normal P-Plot for Rating

Hypothesis:

H_0 = The rating are normally distributed.

H_1= The rating are not normally distributed.

$\alpha = 0.05$

$p - value = 0.0001$

Decision= Reject H_0 since the $p - value\ (0.0001) < \alpha = 0.05$

Conclusion = The rating is not normally distributed.

Interpretation:

Table 6 shows that there is enough evidence to conclude that the customer stratification rating is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 6 which shows that the distribution is not normal in the probability plot of rating.

c. Tax

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 7: Normality Test for Tax

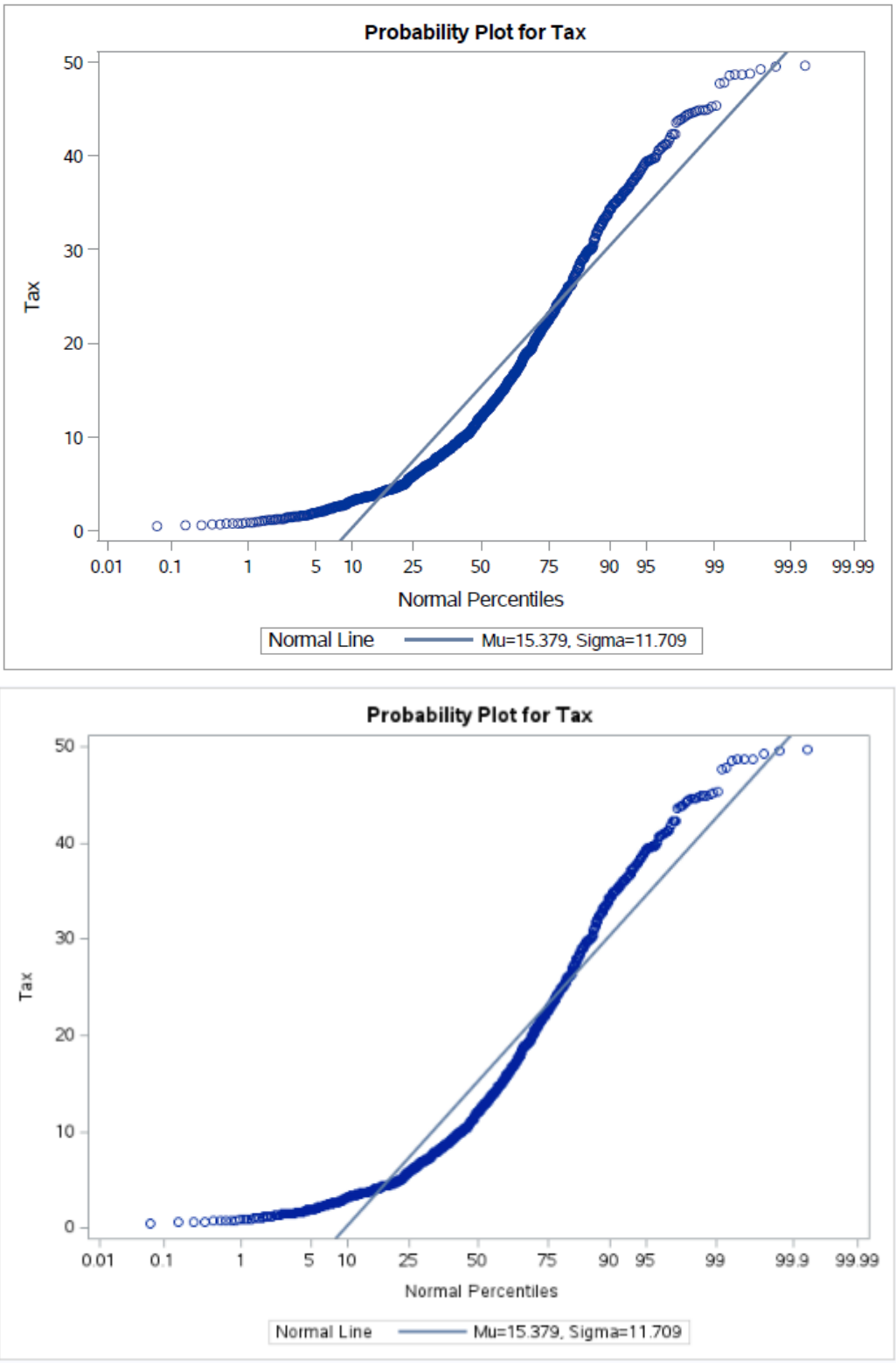


Figure 7: Normal P-Plot for Tax

Hypothesis:

H_0 = The tax are normally distributed.

H_1= The tax are not normally distributed.

$\alpha = 0.05$

$p - value = 0.0001$

Decision= Reject H_0 since the $p - value\ (0.0001) < \ \alpha = 0.05$

Conclusion = The tax is not normally distributed.

Interpretation:

Table 7 shows that there is enough evidence to conclude that the tax fee for customer buying is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 7 which shows that the distribution is not normal in the probability plot of tax.

d. Total

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 8: Normality Test for Total

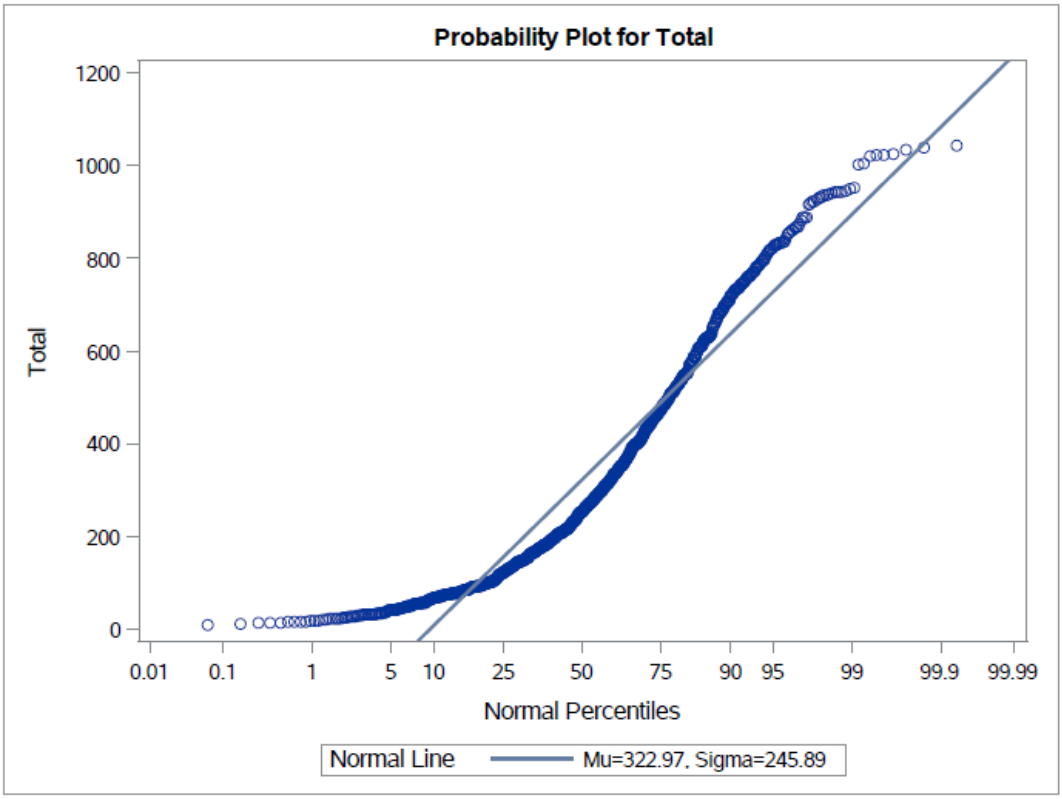


Figure 8: Normal P-Plot for Total

Hypothesis:

H_0 = The total are normally distributed.

H_1= The total are not normally distributed.

$\alpha = 0.05$

$p - value = 0.0001$

Decision= Reject H_0 since the $p - value\ (0.0001) < \alpha = 0.05$

Conclusion = The total is not normally distributed.

Interpretation:

Table 8 shows that there is enough evidence to conclude that total price including tax is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 8 which shows that the distribution is not normal in the probability plot of total.

e. Unit_price

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.951883	$Pr < W$	<0.0001

Table 9: Normality Test for Unit_price

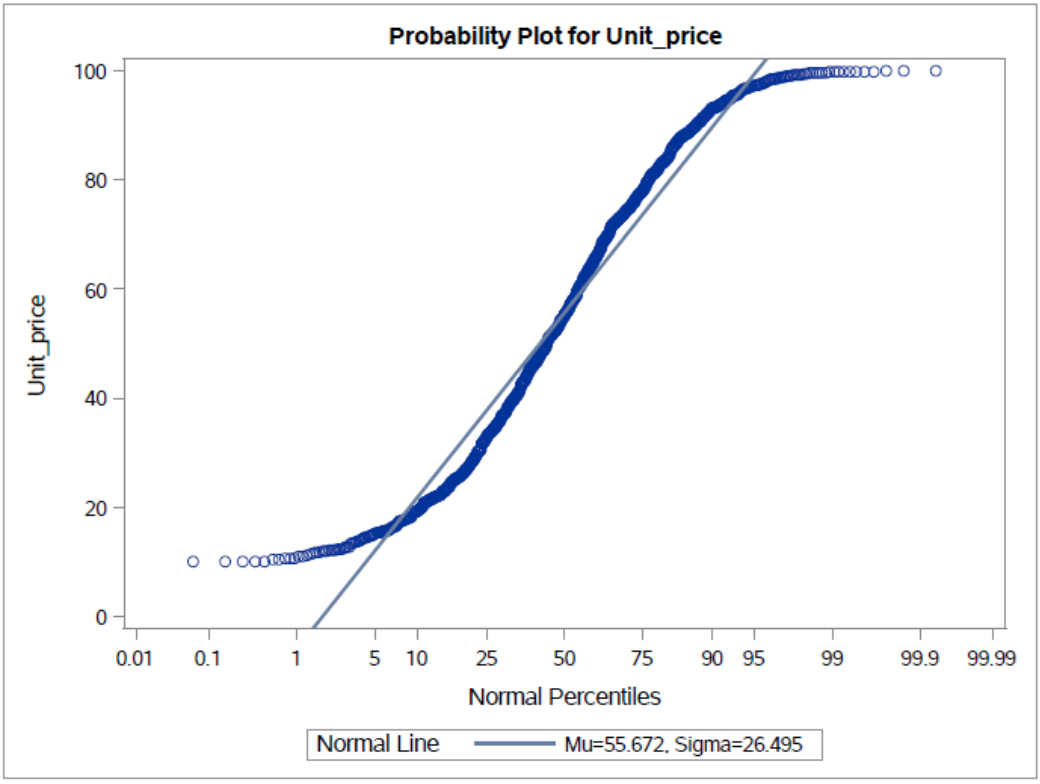


Figure 9: Normal P-Plot for Unit_Price

Hypothesis:

H_0 = The unit price are normally distributed.

H_1= The unit price are not normally distributed.

$\alpha = 0.05$

$p - value = 0.0001$

Decision= Reject H_0 since the $p - value\ (0.0001) < \alpha = 0.05$

Conclusion = The unit price is not normally distributed.

Interpretation:

Table 9 shows that there is enough evidence to conclude that price of each product is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 9 which shows that the distribution is not normal in the probability plot of unit price.

f. COGS

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.932585	$Pr < W$	<0.0001

Table 10: Normality Test for COGS

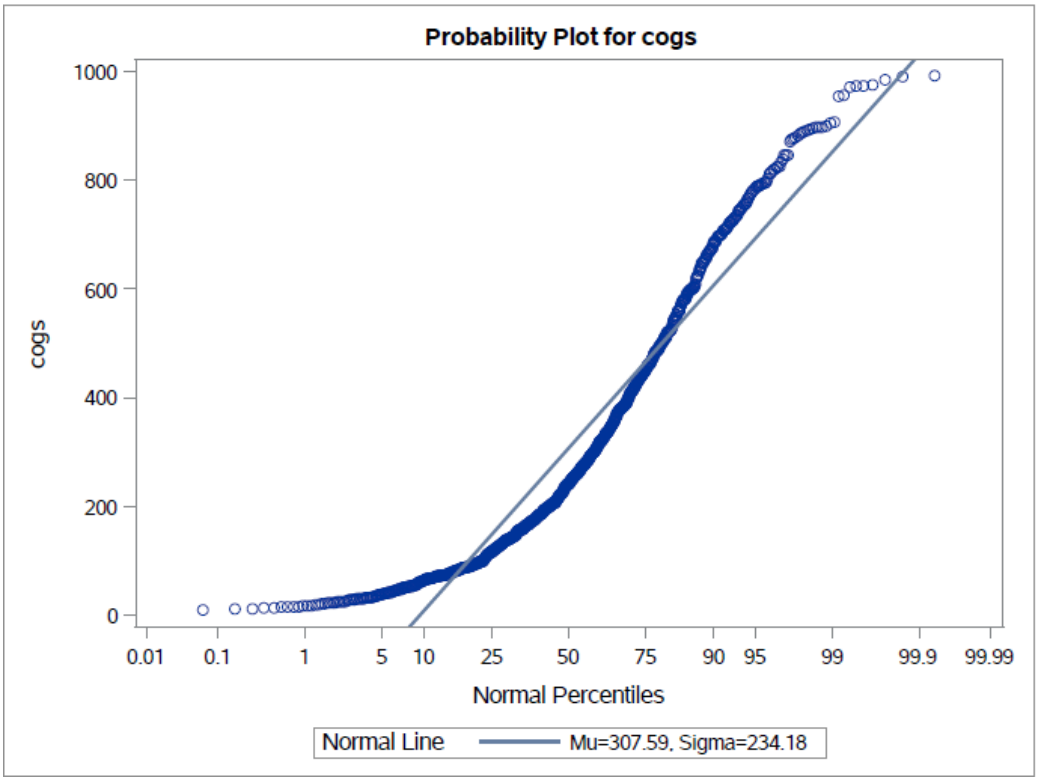


Figure 10: Normal P-Plot for COGS

Hypothesis:

H_0 = The COGS are normally distributed.

H_1= The COGS are not normally distributed.

$\alpha = 0.05$

$p - value = 0.0001$

Decision= Reject H_0 since the $p-value\ (0.0001) < \alpha = 0.05$

Conclusion = The cogs is not normally distributed.

Interpretation:

Table 10 shows that there is enough evidence to conclude that cost of good sold is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 10 which shows that the distribution is not normal in the probability plot of COGS.

g. gross_income

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 11: Normality Test for Gross_Income

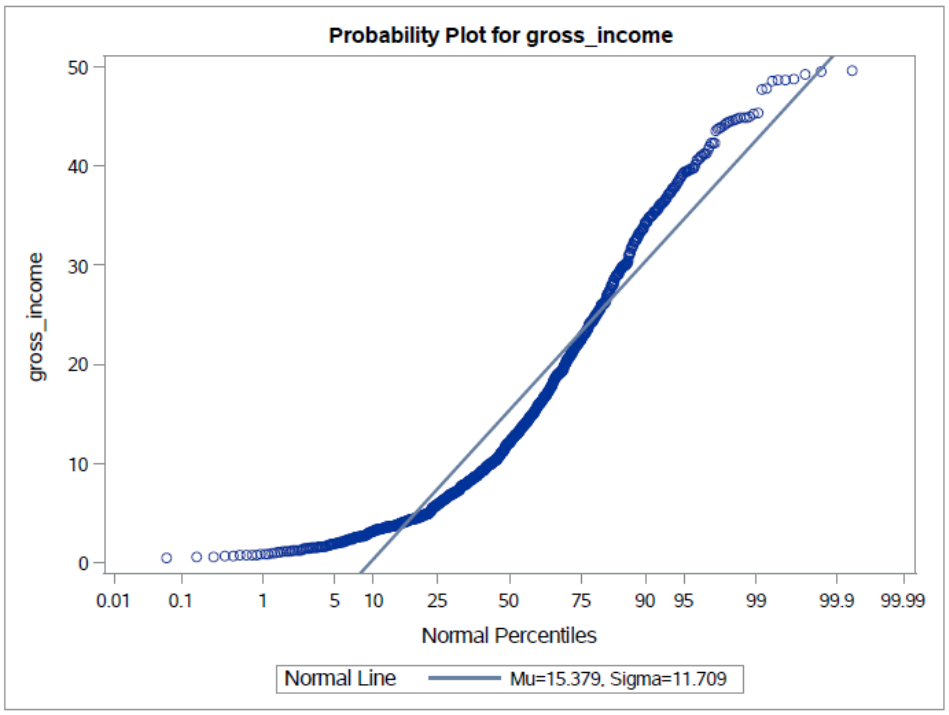


Figure 11: Normal P-Plot for Gross_Income

Hypothesis:

H_0 = The gross income are normally distributed.

H_1= The gross income are not normally distributed.

$\alpha = 0.05$

$p-value = 0.0001$

Decision= Reject H_0 since the $p-value\ (0.0001) < \alpha = 0.05$

Conclusion = The gross income is not normally distributed.

Interpretation:

Table 11 shows that there is enough evidence to conclude that gross income is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). This can also be proven by Figure 11 which shows that the distribution is not normal in the probability plot of gross income.

3.3.2 Multiple Linear Regression

Refer Appendix (11) for the SAS code.

The use of this method is to identify the relationship between total sales (dependent variable) and multiple independent variables which include branch, gender and product line. It can also help to determine whether the independent variables are significant to total sales and develop a model to predict the total sales. Appendix 10 refers to the SAS code of the test.

Source	DF	Sum of Square	Mean Square	F Value	Pr>F
Model	3	268338	89446	1.48	0.2180
Error	996	60130800	60372		
Corrected Total	999	60399138			

Table 12: ANOVA Test on Model Validity

Interpretation: The regression model explains a significant portion of the variation in the dependent variable, as indicated by the high F value and the P-value is 0.2180, which is greater than $\alpha = 0.05$ .We fail to reject the null hypothesis, h_0 that all the regression coefficients are equal to zero. The sum of squares for the model indicates that a substantial amount of variation is explained by the independent variables included in the model. The degrees of freedom and mean squares provide further details on the distribution of variance. Overall, the model is not statistically significant, and the predictors do not have a meaningful relationship with the dependent variable.

Root MSE	245.70773	R-Square	0.0044
Dependent Mean	322.96675	Adjusted R-Square	0.0014
Coefficient Variance	76.07834		

Table 13: Model Fit Statistics

Interpretation: Based on Table 13, the root MSE is 245.70773 represents the standard deviation of the residuals. The value of R-Square is 0.0044 indicates that approximately 0.44% of the variance in total sales can be explained by branch, gender and product line factors while the remaining 99.56% is explained by other factors. In other words, these values indicate that the model is not a good fit for the data and the predictors have a very weak influence on the total sales.

Variable	Label	DF	Parameter Estimate	Standard Error	t-Value	Pr > ltl
Intercept	Intercept	1	307.89614	17.93867	17.16	<0.0001
Branch	Branch	1	11.74592	9.52694	1.23	0.2179
Gender	Gender	1	23.97457	15.59704	1.54	0.1246
Product_line	Product_line	1	-3.31998	4.56480	-0.73	0.4672

Table 14: Parameter Estimates for Multiple Linear Regression

Interpretation: The Table 14 provides the results of a multiple linear regression analysis for predicting total sales based on four variables: Branch, Gender and Product Line. For the intercept, the baseline value of total sales when all predictor variables are zero. The intercept is statistically significant (P-value < 0.0001). The branch is not statistically significant (P-value = 0.2179), indicating that the variable does not have significant effect on total sales. Next, gender also not statistically significant (P-value = 0.1246), indicating that gender does not have a significant effect on total sales. For Product Line is not statistically significant (P-value = 0.4672), it indicates that product line also does not have a significant effect on total sales.

The estimated Multiple Linear Regression (MLR) equation based on these results is:

$$\overline{Y} = 307.89614 + 11.7459x_1 + 23.97457x_2 - 3.31998x_3$$

Where:

$\overline{Y}$ = Total Sales

x_1= = Branch

x_2= = Gender

x_3= = Product Line

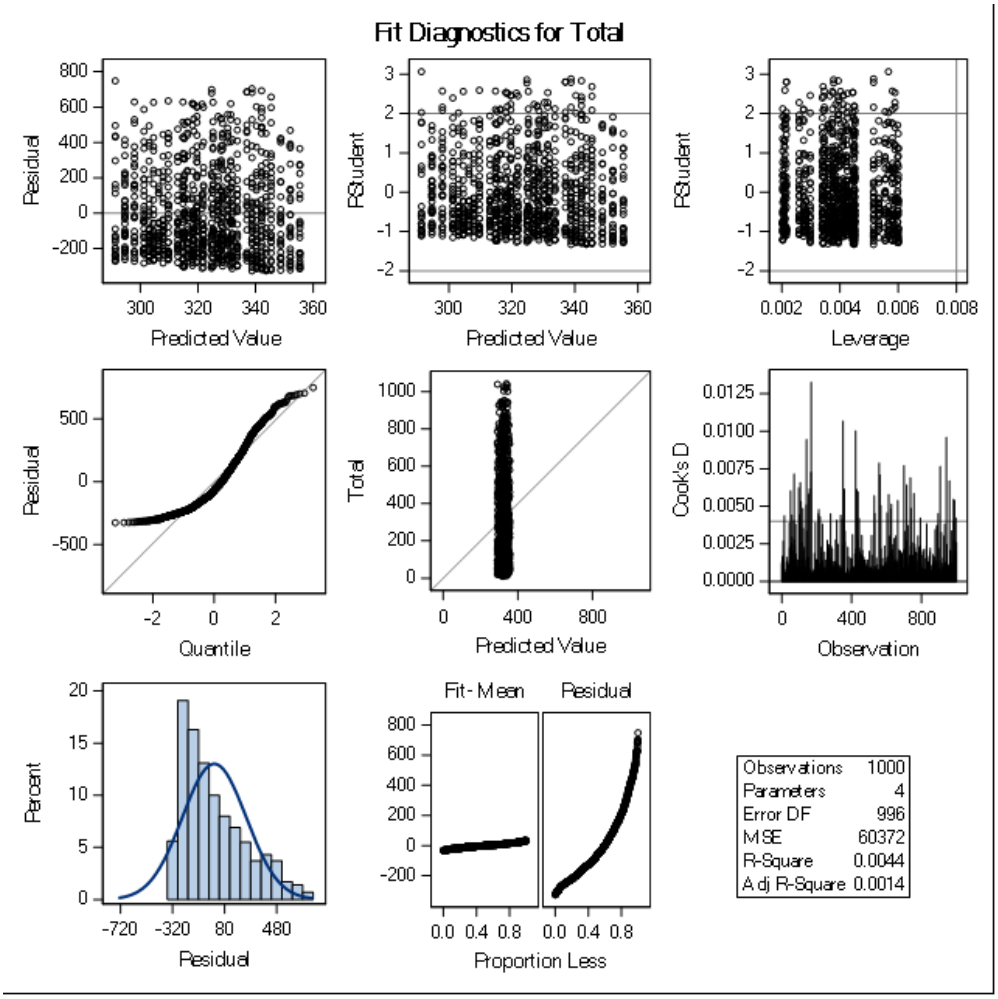


Figure 12: Diagnostics for Total Sales

Interpretation: Based on Figure 12, it shows a series of fit diagnostic plots for a regression model predicting total sales. The residuals vs predicted value are scattered zero, indicating that the model does not have a systematic relationship. Here, there is a slight pattern, suggesting possible problems with the model's assumptions. The Q-Q plot residuals compare the distribution of residuals to a normal distribution. Ideally, the points should fall along the diagonal line, indicating normality. The plot shows some deviation from normality, particularly in the tails, suggesting potential issues with the model's assumptions. The Cook's Distance plot highlights observations with high Cook's distances, indicating a strong influence on the model's coefficients. While there are a few observations with relatively high Cook's distances, they do not seem to be excessively influential. The histogram of residuals appears symmetrical and centred around zero, which is a positive sign. However, the deviation from normality observed in the Q-Q plot suggests some potential issues. The R-squared value indicates that approximately 0.44% of the variability in total sales is explained by the model. This is a very weak fit, and the model might not be a useful predictor of total sales.

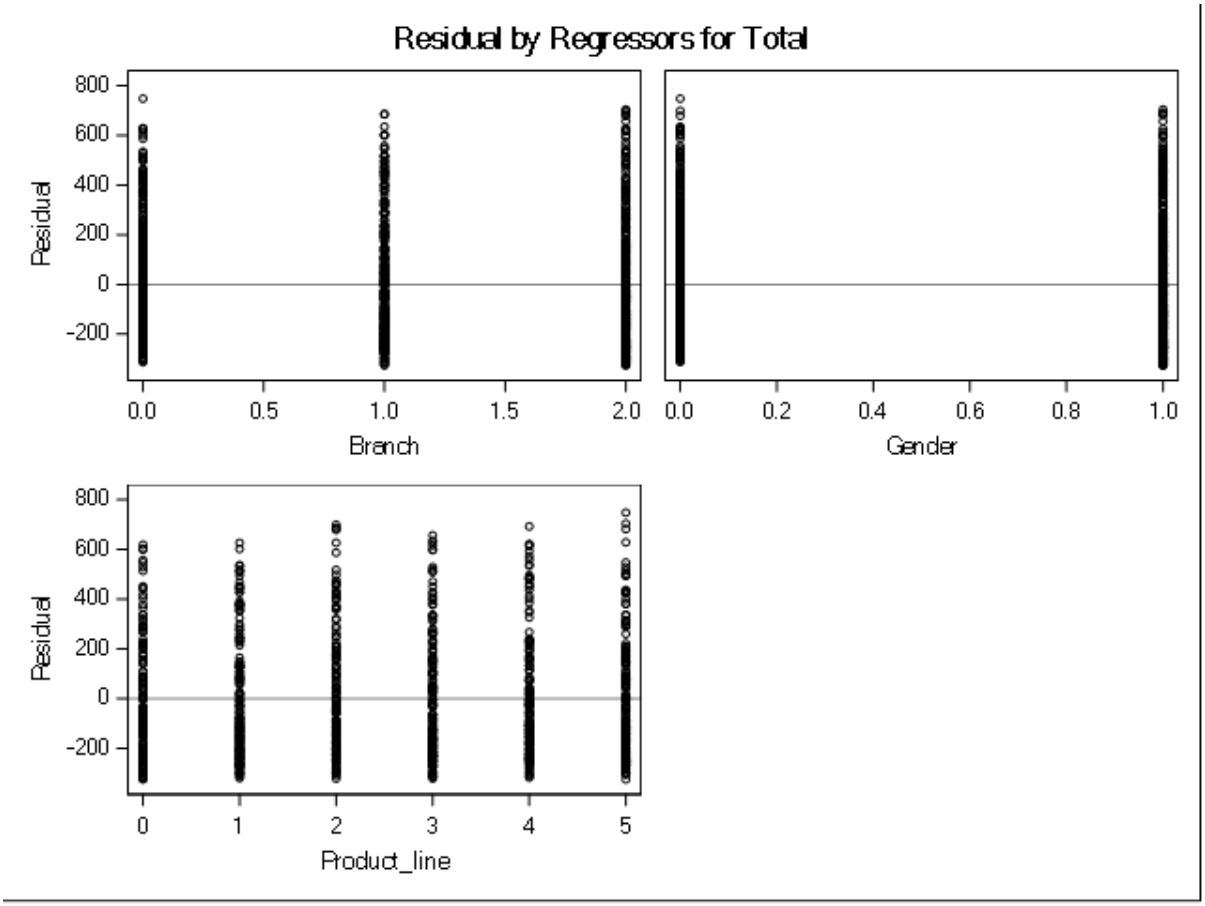


Figure 13: Plot of Residual vs Independent Variables

Interpretation: Based on Figure 13, it shows residual plots for four different regressors predicting Total Sales. For Branch, the residuals are not spread evenly across the different branches. The residuals are concentrated in the Branch A and the Branch B, with very few residuals in the Branch C. This suggests that the model might not be accurately capturing the relationship between branch and the dependent variable for the Branch C. The Gender variable the residuals are not spread evenly across genders. Most of the residuals are clustered around male and female categories. There appears to be a slight pattern that female observations have more negative residuals while male observations have more positive residuals. For Product Line variable, there is a clear pattern in the residuals for each product line. The residuals are not evenly spread across the five product lines. This suggests that the model might not be accurately capturing the relationship between product line and the dependent variable. For example, residuals are clustered around product line 2, suggesting that the model might be underestimating the effect of product line 2 on the dependent variable. Overall, the residual plots for each variable show that there is a clear pattern in the residuals. This suggests that the model might not be accurately capturing the relationship between these independent variables and the dependent variable.

3.3.3 Wilcoxon Rank Sum Test

Refer Appendix (8) for the SAS code to check the normality test before decided to use which test either Independent t test or Wilcoxon Rank Sum Test.

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 15: Normality test for Gender = Female

Table 15 shows that there is enough evidence to conclude that gender = female is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05)

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 16: Normality test for Gender = Male

Table 16 shows that there is enough evidence to conclude that gender = Male is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05)

Refer **Appendix (8)** for the SAS code.

Wilcoxon Scores (Rank Sums) for Variable Gross_income Classified by Variable Gender					
Gender	**N**	**Sum of Scores**	**Expected Under H0**	**Std Dev Under H0**	**Mean Score**
Female	501	258248.50	250750.50	4566.62698	515.466068
Male	499	242251.50	249749.50	4566.62698	485.473948
Average scores were used for ties.					

Table 17: Wilcoxon Scores (Rank Sums) for Variable Gross_income Classified by Variable Gender

Wilcoxon Two-Sample Test					
Statistic	**Z**	**Pr < Z**	**Pr > |Z|**	**T Approximation**	
				Pr < Z	**Pr > |Z|**
242251.5	-1.6418	0.0503	0.1006	0.0505	0.1009
Z includes a continuity correction of 0.5.					

Table 18: Wilcoxon Two-Sample Test

Table 18 shows that the result indicates that there is insufficient evidence to claim that there is a significant difference in gross income between genders female or male since the p-value (0.1006) is more than the alpha value (0.05).

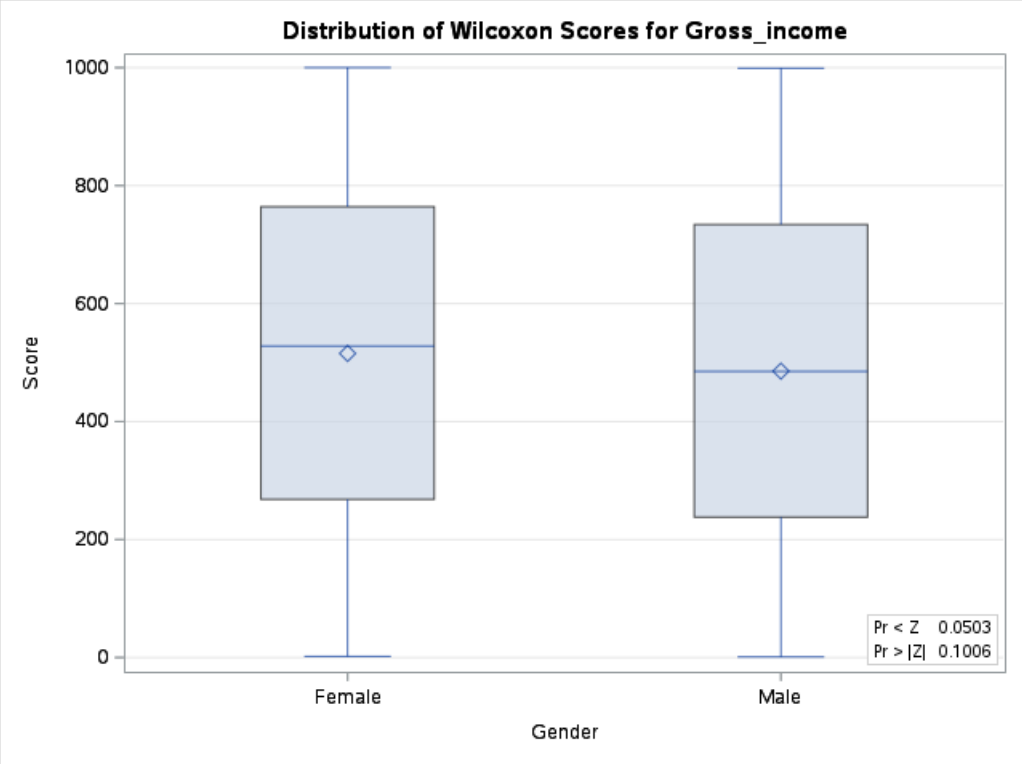


Figure 14: Box Plot of Wilcoxon Scores for Gross Income by Gender

Based on the Figure 14, the two-sided p-value (Pr > |Z| = 0.1006) is greater than the common significance level of 0.05. This indicates that the difference in Wilcoxon scores between males and females is not statistically significant. The Wilcoxon rank-sum test does not provide sufficient evidence to conclude a significant difference in gross_income between males and females. Both groups have similar distributions of gross income based on this analysis.

3.3.4 Kruskal-Wallis Test

Refer Appendix (9) for SAS code to check the normality of each product lines before decided to use which test either Analysis of Variance (ANOVA) or Kruskal-Wallis test.

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	< 0.0001

Table 19: Normality Test for Gross_Income

Table 19 shows that there is enough evidence to conclude that gross income is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Normality		
Test	Statistic	P Value

Shapiro-Wilk	W	0.893622	$Pr < W$	< 0.0001

Table 20: Normality test for Product Line= Electronic Accessories

Table 20 shows that there is enough evidence to conclude that product line = Electronic Accessories is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.904107	$Pr < W$	< 0.0001

Table 21: Normality test for Product Line= Fashion Accessories

Table 21 shows that there is enough evidence to conclude that Product Line= Fashion Accessories is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.889118	$Pr < W$	< 0.0001

Table 22: Normality test for Product Line= Food and Beverages

Table 22 shows that there is enough evidence to conclude that Product Line= Food and Beverages is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.921659	$Pr < W$	< 0.0001

Table 23: Normality test for Product Line= Health and Beauty

Table 23 shows that there is enough evidence to conclude that Product Line= Health and Beauty is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.903188	$Pr < W$	< 0.0001

Table 24: Normality test for Product Line= Home and Lifestyle

Table 24 shows that there is enough evidence to conclude that Product Line= Home and Lifestyle is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.917642	$Pr < W$	< 0.0001

Table 25: Normality test for Product Line= Sports and Travel

Table 25 shows that there is enough evidence to conclude that Product Line= Sports and Travel is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Wilcoxon Scores (Rank Sums) for Variable gross_income Classified by Variable Product_line					
Product_line	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Health and Beauty	152	77058.00	76076.0	3279.03553	506.960526
Electronic Accessories	170	84016.50	85085.0	3430.75662	494.214706
Home and Lifestyle	160	82652.00	80080.0	3348.31291	516.575000
Sports and Travel	166	84787.00	83083.0	3398.31374	510.765060
Food and Beverages	174	87273.00	87087.0	3462.51011	501.568966
Fashion Accessories	178	84713.50	89089.0	3493.59303	475.918539
Average scores were used for ties.					

Table 26: Wilcoxon Scores (Rank Sums) for Variable gross_income Classified by Variable Product_line

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
2.1537	5	0.8275

Table 27: Kruskal-Wallis Test

Table 27 shows that the result indicates that there is insufficient evidence to claim that there is a significant difference in gross income between all product lines since the p-value (0.8275) is more than the alpha value (0.05). Therefore, there is no significant difference in gross income between all product lines.

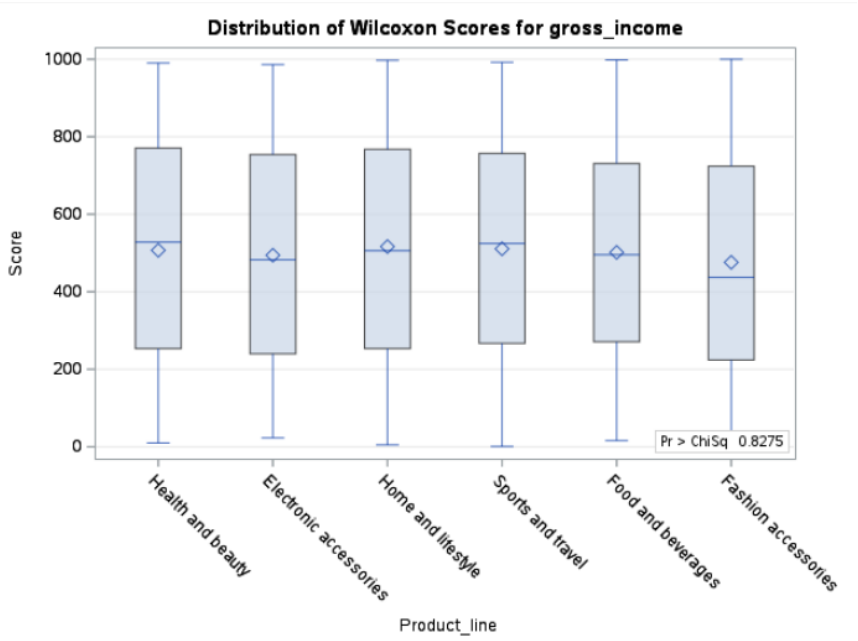


Figure 15: Box Plot of Wilcoxon Scores for Gross Income by Product Line

Based on the Kruskal-Wallis test, it shows that there is no sufficient evidence to claim that there is a significant difference in gross income across the six product lines: Health and Beauty, Electronic Accessories, Home and Lifestyle, Sports and Travel, Food and Beverages, and Fashion Accessories. The p-value (0.8275) is significantly greater than value of alpha (0.05), we failed to reject the null hypothesis. The boxplots in figure 3.15 supports this conclusion as the median and interquartile ranges of the Wilcoxon scores are shown at similar places for each product line. This shows that there is a consistent pattern in the gross income distribution. In conclusion, the Kruskal-Wallis test shows that gross income does not differ across the six product lines subsequently proves that all product lines perform the same in terms of gross income.

3.3.5 One Sample Median Test

Refer Appendix (12) for the SAS code.

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.958201	$Pr < W$	<0.0001

Table 28: Normality Test for Rating

Table 28 shows that there is enough evidence to conclude that rating is not normally distributed since the probability of the Shapiro-Wilk (<0.0001) is less than alpha value (0.05).

Test for Location:Mu0=5			
Test	Statistic	P	Value

Student's t	t	36.29871	Pr > |t|	<.0001
Sign	M	336.5	Pr >= |M|	<.0001
Signed Rank	S	215515.5	Pr >= |S|	<.0001

Table 29: One Sample Median Test

Based on the table 29, the p-value (<0.0001) is less than alpha value (0.05). This conclude that there is sufficient evidence to indicate that the median rating is not equal to 5.

3.3.6 Spearman Correlation Coefficient

Refer Appendix (10) for the SAS code to check the normality test before decided to use which test either Pearson's Correlation Coefficient or Spearman's Rank Correlation Coefficient.

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 30: Normality test for Cost of Goods Sold

Table 30 shows that there is enough evidence to conclude that the cost of goods sold is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05).

Test for Normality				
Test	Statistic		P Value	
Shapiro-Wilk	W	0.90876	$Pr < W$	<0.0001

Table 31: Normality test for Total

Table 31 shows that there is enough evidence to conclude that the total of sale is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05).

Spearman Correlation Coefficients,N=1000 Prob > $\|r\|$ under H0: Rh0 = 0	
	Total
Cogs	1.00000
Cost of Goods Solds	< .0001

Table 32: Spearman's Rank Correlation Coefficient Test

Based on the Table 32 shows that the Spearman Correlation between Cost of Goods Sold and Total Sales is 1.0000 which suggests a perfect positive rank-order relationship. More than that, the p-value (<0.0001) is extremely small, indicating the correlation is statistically significant.

This means the likelihood of observing this correlation by random chance under the null hypothesis which is no correlation is almost zero.

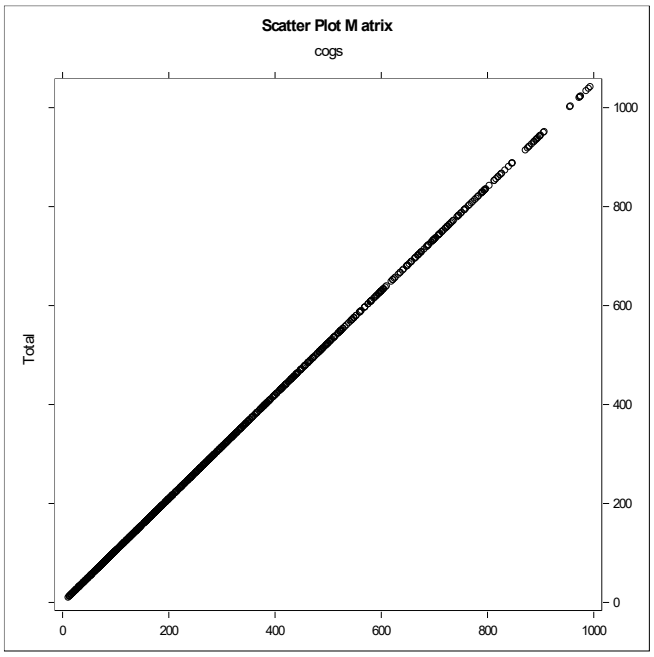


Figure 16: Scatter Plot Matrix of Costs of Goods Sold (COGS)

Moreover, the perfectly positive relationship can be shown through scatter plot. The scatter plot shows a perfectly monotonic relationship. Monotonicity means that as one variable increases, the other variable consistently increases or decreases. In this case, as Cost of Goods Sold $(X - axis)$ increases, Total Sales $(Y - axis)$ increases proportionally without deviation. The data points lie on a straight line, suggesting a perfect linear relationship.

Interpretation: There is statistically significant strong positive monotonic relationship between Cost of Goods Sold and Total Sales. As the Total Sales increases, the cost of goods sold also tend to increase, and this relationship is unlikely to be due to chance given the very low p-value (<.0001)

CHAPTER 4: CONCLUSION AND DISCUSSION

In conclusion, the objectives stated have been achieved to analyse the supermarket sales with the help of SAS Studio.

Multiple Linear Regression revealed all variables have significant effect on total sales. The branch, gender and product line variables does not have effect on total sales as there are the weak relationship between the variables and total sales.

Furthermore, the Wilcoxon Rank Sum test is used to achieve the second objective which is to determine whether there is differs in gross income between gender of female and male. Based on all the results, there is enough evidence to conclude that gender = female is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). Meanwhile, for the second group it shows that there is also enough evidence to conclude that gender = male is not normally distributed since the probability of the Shapiro-Wilk (0.0001) is less than alpha value (0.05). Thus, there is insufficient evidence to claim that there is a significant difference in gross income between genders female or male since the p-value (0.1006) is more than the alpha value (0.05).

Next, Kruskal-Wallis test is used for the third objective, to determine whether if there is no significant difference in the average gross income across different product lines. The result shows that there is no sufficient evidence to claim that there is a significant difference in gross income across the six product lines: Health and Beauty, Electronic Accessories, Home and Lifestyle, Sports and Travel, Food and Beverages, and Fashion Accessories. The p-value (0.8275) is significantly greater than value of alpha (0.05), we failed to reject the null hypothesis. The boxplots from Kruskal-Wallis test also supports this conclusion as the median and interquartile ranges of the Wilcoxon scores are shown at similar places for each product line. This shows that there is a consistent pattern in the gross income distribution. Thus, the Kruskal-Wallis test shows that gross income does not differ across the six product lines subsequently proves that all product lines perform the same in terms of gross income.

Moreover, we use one sample median test as the fourth objective. The test is used to determine if the median rating is equal to 5. The p-value (<0.0001) is less than alpha value (0.05). This conclude that there is sufficient evidence to indicate that the median rating is not equal to 5.

By using inferential statistics, we found that our data was not normally distributed, necessitating the use of non-parametric tests. The Spearman's Rank Correlation Coefficient indicated significant difference between cost of goods sold and total sales. The result shows that there is strong positive relationship between cost of goods sold and total sales.

CHAPTER 5: REFERENCES

Supermarket sales. (2019, May 27). https://www.kaggle.com/datasets/aungpyaeap/supermarket-sales?resource=download

Kotler, P., & Keller, K. L. (2016). *Marketing Management* (15th ed.). Pearson. https://www.pearson.com/en-us/subject-catalog/p/marketing-management/P200000007478/9780134236933

Zeithaml, V. A., Berry, L. L., & Parasuraman, A. (1996). *The Behavioral Consequences of Service Quality*. *Journal of Marketing*, 60(2), 31–46. https://www.researchgate.net/publication/329258469_The_Behavioral_Consequences_of_Service_Quality

Chen, J., & Wang, X. (2020). *The Impact of COVID-19 on Consumer Shopping Behavior: Evidence from a Food-Based Retailer*. *Journal of Retailing and Consumer Services*, 59, 102393. https://pmc.ncbi.nlm.nih.gov/articles/PMC9485879/

CHAPTER 6: APPENDICES

Appendix 1: Assigning a Libref

```
libname orion '/home/u64051469/ASSIGNMENT 3';
```

Appendix 2: Reading Excel Spreadsheets

```
proc import out=orion.smsales
datafile="/home/u64051469/ASSIGNMENT 3/supermarket sales.xlsx"
DBMS=XLSX;
GETNAMES=YES;
run;
```

Appendix 3: The Descriptor Portion

```
proc contents data=orion.smsales;
run;
```

The CONTENTS Procedure

Data Set Name	ORION.SMSALES	**Observations**	1000
Member Type	DATA	**Variables**	13
Engine	V9	**Indexes**	0
Created	01/24/2025 04:24:43	**Observation Length**	120
Last Modified	01/24/2025 04:24:43	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1090
Obs in First Data Page	1000
Number of Data Set Repairs	0
Filename	/home/u64051473/GROUP PROJECT/smsales.sas7bdat
Release Created	9.0401M7
Host Created	Linux
Inode Number	7968068930
Access Permission	rw-r--r--
Owner Name	u64051473
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
2	Branch	Char	1	$1.	$1.	Branch
3	City	Char	9	$9.	$9.	City
4	Gender	Char	6	$6.	$6.	Gender
1	Invoice_ID	Char	11	$11.	$11.	Invoice_ID
10	Payment	Char	11	$11.	$11.	Payment
5	Product_line	Char	22	$22.	$22.	Product_line
7	Quantity	Num	8	BEST.		Quantity
13	Rating	Num	8	BEST.		Rating
8	Tax	Num	8	BEST.		Tax
9	Total	Num	8	BEST.		Total
6	Unit_price	Num	8	BEST.		Unit_price
11	cogs	Num	8	BEST.		cogs
12	gross_income	Num	8	BEST.		gross_income

Appendix 4: Display the partial dataset

```
proc print data= orion.smsales;
run;
```

Obs	Invoice_ID	Branch	City	Gender	Product_line	Unit_price	Quantity	Tax	Total	Payment	cogs	gross_income	Rating
1	750-67-8428	A	Yangon	Female	Health and beauty	74.69	7	26.1415	548.9715	Ewallet	522.83	26.1415	9.1
2	226-31-3081	C	Naypyitaw	Female	Electronic accessories	15.28	5	3.82	80.22	Cash	76.4	3.82	9.6
3	631-41-3108	A	Yangon	Male	Home and lifestyle	46.33	7	16.2155	340.5255	Credit card	324.31	16.2155	7.4
4	123-19-1176	A	Yangon	Male	Health and beauty	58.22	8	23.288	489.048	Ewallet	465.76	23.288	8.4
5	373-73-7910	A	Yangon	Male	Sports and travel	86.31	7	30.2085	634.3785	Ewallet	604.17	30.2085	5.3
6	699-14-3026	C	Naypyitaw	Male	Electronic accessories	85.39	7	29.8865	627.6165	Ewallet	597.73	29.8865	4.1
7	355-53-5943	A	Yangon	Female	Electronic accessories	68.84	6	20.652	433.692	Ewallet	413.04	20.652	5.8
8	315-22-5665	C	Naypyitaw	Female	Home and lifestyle	73.56	10	36.78	772.38	Ewallet	735.6	36.78	8
9	665-32-9167	A	Yangon	Female	Health and beauty	36.26	2	3.626	76.146	Credit card	72.52	3.626	7.2
10	692-92-5582	B	Mandalay	Female	Food and beverages	54.84	3	8.226	172.746	Credit card	164.52	8.226	5.9
11	351-62-0822	B	Mandalay	Female	Fashion accessories	14.48	4	2.896	60.816	Ewallet	57.92	2.896	4.5
12	529-56-3974	B	Mandalay	Male	Electronic accessories	25.51	4	5.102	107.142	Cash	102.04	5.102	6.8
13	365-64-0515	A	Yangon	Female	Electronic accessories	46.95	5	11.7375	246.4875	Ewallet	234.75	11.7375	7.1
14	252-56-2699	A	Yangon	Male	Food and beverages	43.19	10	21.595	453.495	Ewallet	431.9	21.595	8.2
15	829-34-3910	A	Yangon	Female	Health and beauty	71.38	10	35.69	749.49	Cash	713.8	35.69	5.7
16	299-46-1805	B	Mandalay	Female	Sports and travel	93.72	6	28.116	590.436	Cash	562.32	28.116	4.5
17	656-95-9349	A	Yangon	Female	Health and beauty	68.93	7	24.1255	506.6355	Credit card	482.51	24.1255	4.6
18	765-26-6951	A	Yangon	Male	Sports and travel	72.61	6	21.783	457.443	Credit card	435.66	21.783	6.9
19	329-62-1586	A	Yangon	Male	Food and beverages	54.67	3	8.2005	172.2105	Credit card	164.01	8.2005	8.6
20	319-50-3348	B	Mandalay	Female	Home and lifestyle	40.3	2	4.03	84.63	Ewallet	80.6	4.03	4.4
21	300-71-4605	C	Naypyitaw	Male	Electronic accessories	86.04	5	21.51	451.71	Ewallet	430.2	21.51	4.8
22	371-85-5789	B	Mandalay	Male	Health and beauty	87.98	3	13.197	277.137	Ewallet	263.94	13.197	5.1
23	273-16-6619	B	Mandalay	Male	Home and lifestyle	33.2	2	3.32	69.72	Credit card	66.4	3.32	4.4
24	636-48-8204	A	Yangon	Male	Electronic accessories	34.56	5	8.64	181.44	Ewallet	172.8	8.64	9.9
25	549-59-1358	A	Yangon	Male	Sports and travel	88.63	3	13.2945	279.1845	Ewallet	265.89	13.2945	6
26	227-03-5010	A	Yangon	Female	Home and lifestyle	52.59	8	21.036	441.756	Credit card	420.72	21.036	8.5
27	649-29-6775	B	Mandalay	Male	Fashion accessories	33.52	1	1.676	35.196	Cash	33.52	1.676	6.7

Appendix 5: Summary Statistics on 4 quantitative variables with the output

```
proc means data=orion.smsales maxdec=4;
var Unit_price Quantity Total gross_income;
run;
```

The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
Unit_price	Unit_price	1000	55.6721	26.4946	10.0800	99.9600
Quantity	Quantity	1000	5.5100	2.9234	1.0000	10.0000
Total	Total	1000	322.9667	245.8853	10.6785	1042.6500
gross_income	gross_income	1000	15.3794	11.7088	0.5085	49.6500

Appendix 6: One-Way Frequency Report

```
proc freq data=orion.smsales;
table Product_line;
run;
```

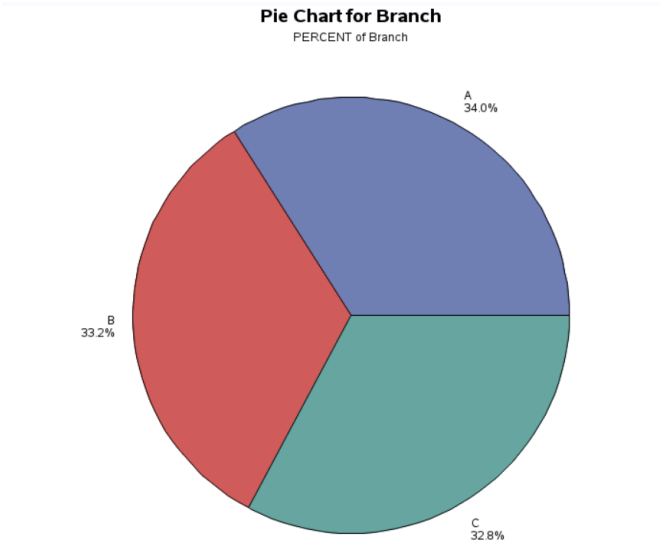
The FREQ Procedure

Product_line				
Product_line	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Electronic accessories	170	17.00	170	17.00
Fashion accessories	178	17.80	348	34.80
Food and beverages	174	17.40	522	52.20
Health and beauty	152	15.20	674	67.40
Home and lifestyle	160	16.00	834	83.40
Sports and travel	166	16.60	1000	100.00

Appendix 7: Pie Chart

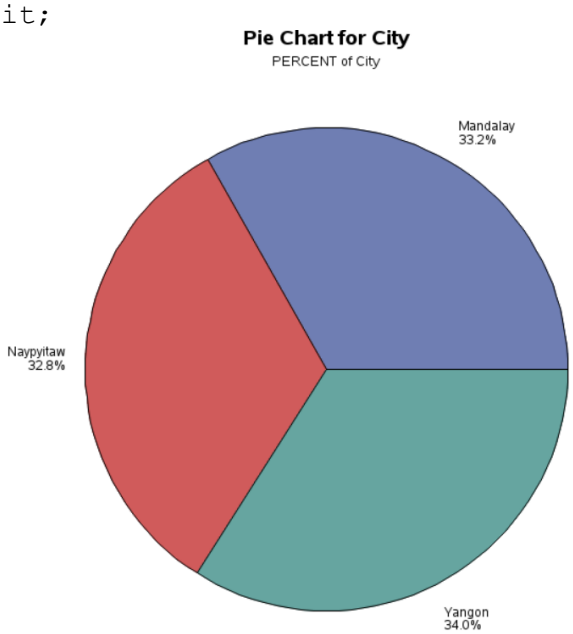
Branch

```
title "Pie Chart for Branch";
proc gchart data=orion.smsales;
                    pie Branch/type=percent;
run;
quit;
```



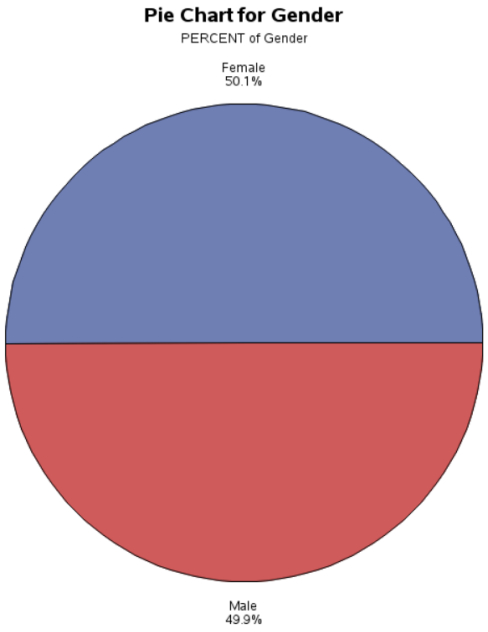
City

```
title "Pie Chart for City";
proc gchart data=orion.smsales;
                    pie City/type=percent;
run;
quit;
```



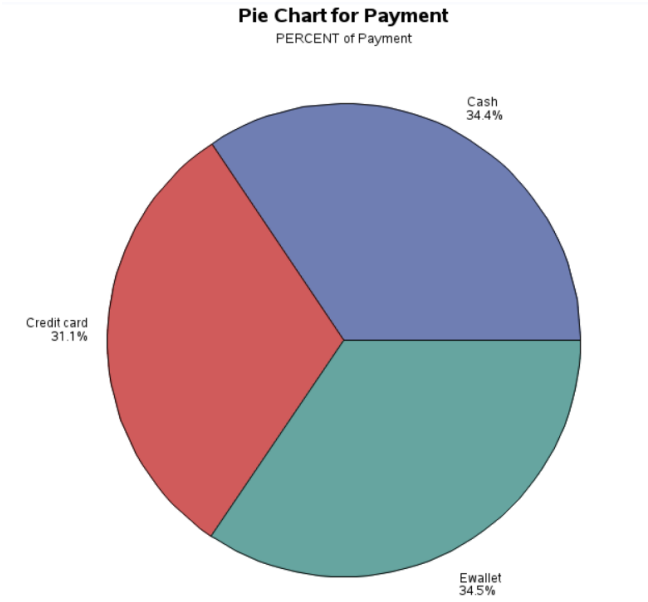
Gender

```
title "Pie Chart for Gender";
proc gchart data=orion.smsales;
                    pie Gender/type=percent;
run;
quit;
```



Payment

```
title "Pie Chart for Payment";
proc gchart data=orion.smsales;
                    pie Payment/type=percent;
run;
quit;
```



Appendix 8: Wilcoxon Rank Sum Test

```
proc univariate data=orion.smsales normal;
   class gender;
```

```
   var gross_income;
   probplot / normal (mu=est sigma=est);
run;
```

Female

Tests for Normality				
Test	**Statistic**		**p Value**	
Shapiro-Wilk	**W**	0.918615	**Pr < W**	<0.0001
Kolmogorov-Smirnov	**D**	0.114233	**Pr > D**	<0.0100
Cramer-von Mises	**W-Sq**	1.921601	**Pr > W-Sq**	<0.0050
Anderson-Darling	**A-Sq**	12.22787	**Pr > A-Sq**	<0.0050

Male

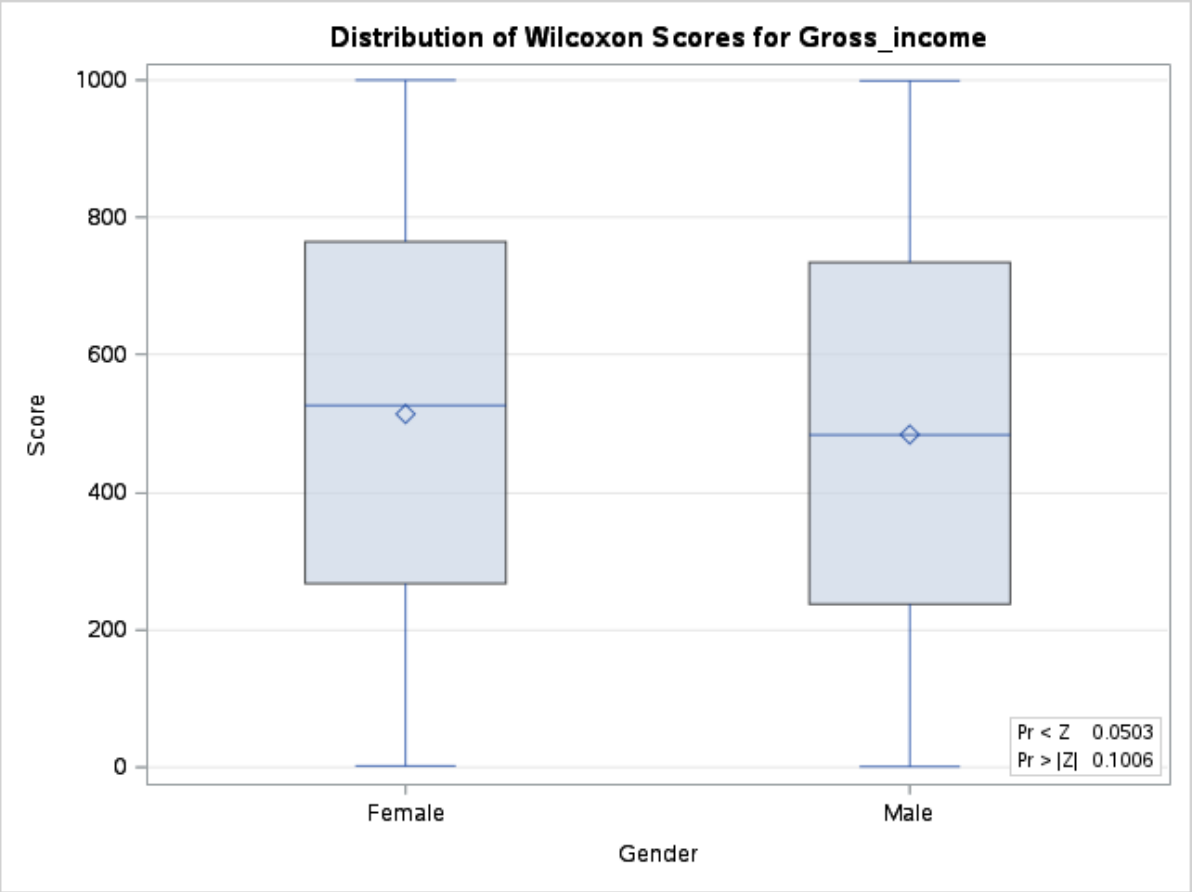
Tests for Normality				
Test	**Statistic**		**p Value**	
Shapiro-Wilk	**W**	0.897232	**Pr < W**	<0.0001
Kolmogorov-Smirnov	**D**	0.135395	**Pr > D**	<0.0100
Cramer-von Mises	**W-Sq**	2.781363	**Pr > W-Sq**	<0.0050
Anderson-Darling	**A-Sq**	16.60531	**Pr > A-Sq**	<0.0050

/*Wilcoxon Rank Sum Test*/

```
title 'Performing a Wilcoxon Rank Sum Test';
proc npar1way data=orion.smsales wilcoxon;
   class gender;
   var gross_income;
run;
title;
```

Wilcoxon Scores (Rank Sums) for Variable Gross_income Classified by Variable Gender					
Gender	**N**	**Sum of Scores**	**Expected Under H0**	**Std Dev Under H0**	**Mean Score**
Female	501	258248.50	250750.50	4566.62698	515.466068
Male	499	242251.50	249749.50	4566.62698	485.473948
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
242251.5	-1.6418	0.0503	0.1006	0.0505	0.1009
Z includes a continuity correction of 0.5.					



Performing a Wilcoxon Rank Sum Test

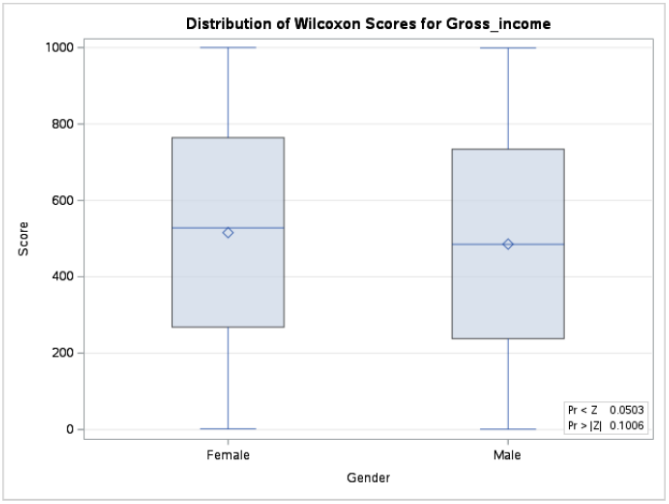
The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Gross_income Classified by Variable Gender					
Gender	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Female	501	258248.50	250750.50	4566.62698	515.466068
Male	499	242251.50	249749.50	4566.62698	485.473948
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
Z includes a continuity correction of 0.5.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
242251.5	-1.6418	0.0503	0.1006	0.0505	0.1009
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
2.6959	1	0.1006



Appendix 9: Kruskal-Wallis Test

*To test the normality for each product line;

```
proc univariate data=orion.smsales normal;
      var gross_income;
      class product_line;
      probplot/normal (mu=est sigma=est);
run;
```

Electronic Accessories

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.893622	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.14321	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.016544	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	6.139021	Pr > A-Sq	<0.0050

Fashion Accessories

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.904107	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.152233	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.815971	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	5.031178	Pr > A-Sq	<0.0050

Food and Beverages

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.889118	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.125668	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.951297	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	6.040366	Pr > A-Sq	<0.0050

Health and Beauty

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.921659	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.130038	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.586547	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	3.563036	Pr > A-Sq	<0.0050

Home and Lifestyle

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.903188	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.132533	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.810518	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	4.927151	Pr > A-Sq	<0.0050

Sports and Travel

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.917642	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.120097	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.644736	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	4.094509	Pr > A-Sq	<0.0050

```
proc npar1way data=orion.smsales wilcoxon;
var gross_income;
class product_line;
run;
```

The NPAR1WAY Procedure

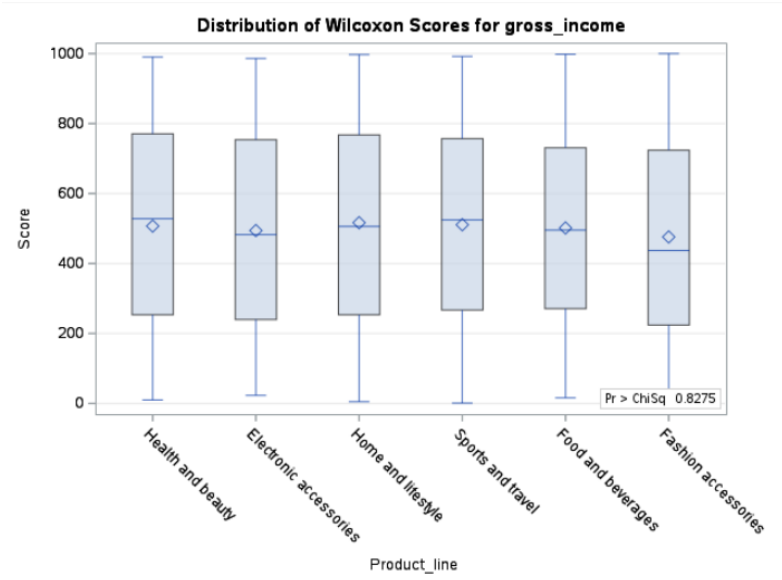
Wilcoxon Scores (Rank Sums) for Variable gross_income
Classified by Variable Product_line

Product_line	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Health and beauty	152	77058.00	76076.0	3279.03553	506.960526
Electronic accessories	170	84016.50	85085.0	3430.75662	494.214706
Home and lifestyle	160	82652.00	80080.0	3348.31291	516.575000
Sports and travel	166	84787.00	83083.0	3398.31374	510.765060
Food and beverages	174	87273.00	87087.0	3462.51011	501.568966
Fashion accessories	178	84713.50	89089.0	3493.59303	475.918539

Average scores were used for ties.

Kruskal-Wallis Test

Chi-Square	DF	Pr > ChiSq
2.1537	5	0.8275



Appendix 10: Spearman's Correlation Coefficient

*To test the normality of COGS and Total

```
proc univariate data= orion.smsales normal;
      var   cogs total;
      probplot/normal (mu=est sigma=est);
run;
```

Cost of Goods Sold

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.90876	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.119161	Pr > D	<0.0100
Cramer-von Mises	W-Sq	4.626547	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	28.47452	Pr > A-Sq	<0.0050

Total

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.90876	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.119161	Pr > D	<0.0100
Cramer-von Mises	W-Sq	4.626547	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	28.47452	Pr > A-Sq	<0.0050

*Spearman Correlation Coefficient Test

```
PROC CORR DATA= ORION.SMSALES spearman
      plots=matrix(histogram);
      label cogs = 'Cost of Goods Solds' total='Total Sales';
      VAR COGS;
      WITH TOTAL;
RUN;
```

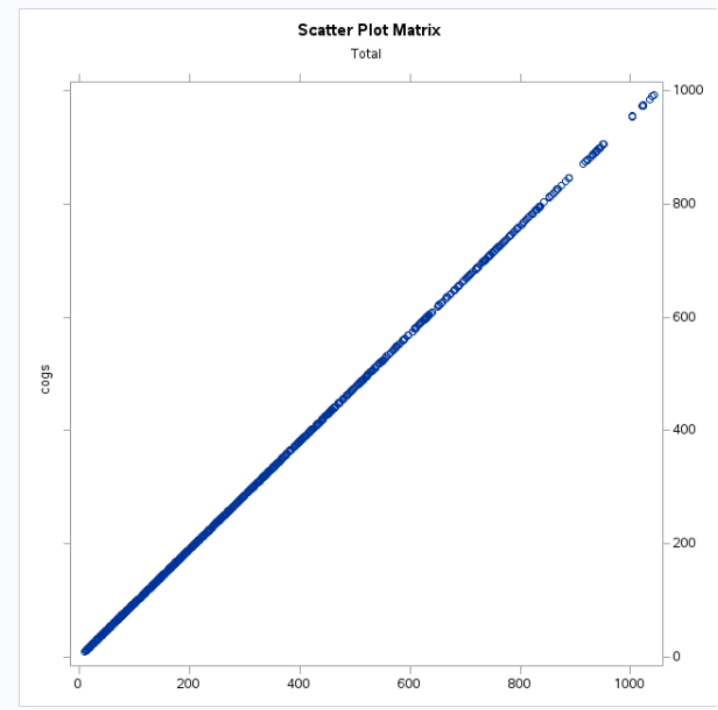
The CORR Procedure

1 With Variables:	cogs
1 Variables:	Total

Simple Statistics							
Variable	N	Mean	Std Dev	Median	Minimum	Maximum	Label
cogs	1000	307.58738	234.17651	241.76000	10.17000	993.00000	Cost of Goods Solds
Total	1000	322.96675	245.88534	253.84800	10.67850	1043	Total Sales

Pearson Correlation Coefficients, N = 1000 Prob > |r| under H0: Rho=0	
	Total
cogs Cost of Goods Solds	1.00000 <.0001

Spearman Correlation Coefficients, N = 1000 Prob > |r| under H0: Rho=0	
	Total
cogs Cost of Goods Solds	1.00000 <.0001



Appendix 11: Multiple Linear Regression

```
proc reg data = orion.smsales;
model Total = Branch Gender Product_line;
output out = resids  p=yhat  r=resid;
run;
quit;
```

The REG Procedure
Model: MODEL1
Dependent Variable: Total Total

Number of Observations Read	1000
Number of Observations Used	1000

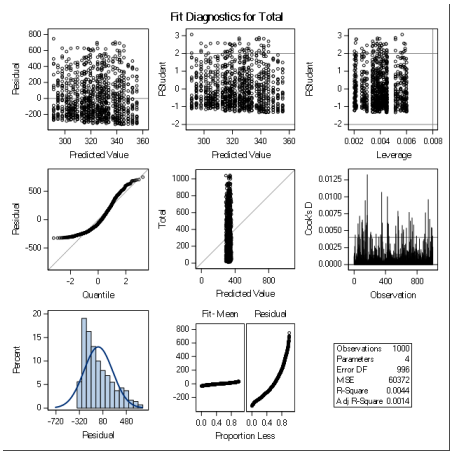
Analysis of Variance

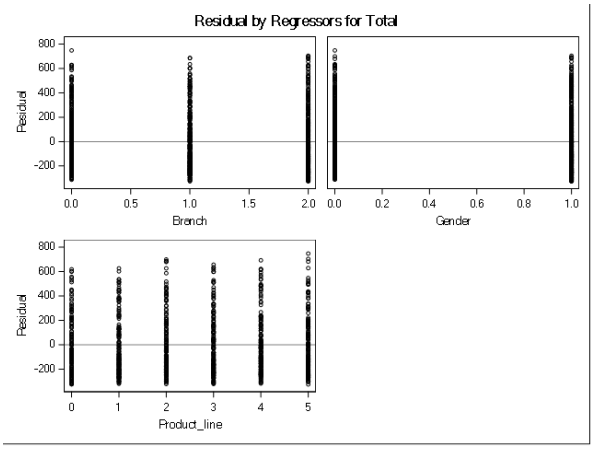
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	268338	89446	1.48	0.2180
Error	996	60130800	60372		
Corrected Total	999	60399138			

Root MSE	245.70773	R-Square	0.0044
Dependent Mean	322.96675	Adj R-Sq	0.0014
Coeff Var	76.07834		

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	307.89614	17.93867	17.16	<.0001
Branch	Branch	1	11.74592	9.52694	1.23	0.2179
Gender	Gender	1	23.97457	15.59704	1.54	0.1246
Product_line	Product_line	1	-3.31998	4.56480	-0.73	0.4672





Appendix 12: One Sample Median Test on Rating with the Output

```
proc univariate data=orion.smsales mu0=5;
      var rating;
run;
```

The UNIVARIATE Procedure
Variable: Rating (Rating)

Moments			
N	1000	Sum Weights	1000
Mean	6.9727	Sum Observations	6972.7
Std Deviation	1.71858029	Variance	2.95351823
Skewness	0.00900965	Kurtosis	-1.1515868
Uncorrected SS	51569.11	Corrected SS	2950.56471
Coeff Variation	24.6472714	Std Error Mean	0.05434628

Basic Statistical Measures			
Location		Variability	
Mean	6.972700	Std Deviation	1.71858
Median	7.000000	Variance	2.95352
Mode	6.000000	Range	6.00000
		Interquartile Range	3.00000

Tests for Location: Mu0=5				
Test	Statistic		p Value	
Student's t	t	36.29871	Pr > |t|	<.0001
Sign	M	336.5	Pr >= |M|	<.0001
Signed Rank	S	215515.5	Pr >= |S|	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	10.00
99%	9.90
95%	9.70
90%	9.40
75% Q3	8.50
50% Median	7.00
25% Q1	5.50
10%	4.50
5%	4.25
1%	4.00
0% Min	4.00

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
4	877	10	61
4	849	10	63
4	791	10	160
4	673	10	388
4	644	10	854